



ODBC Interface Manual



Spillman® Public Safety Software

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Preface

Welcome to the *ODBC Interface User Manual*. This manual is written for Flex users who want to extract data from Flex via a Structured Query Language (SQL) query using the Open Database Connectivity (ODBC) protocol.

The ODBC Interface is available to Flex.

Other manuals

The *RMS User Manual* provides information about the basic features of the Flex software, including how to start and exit Flex, navigate, use screens, search, print, and run reports. The *RMS User Manual* also explains how to use the Hub module, which comprises the tables used by most users.

Windows basics

Before using the software, be familiar with the standard features of Microsoft® Windows®. At a minimum, know how to do the following:

- Use a mouse or keyboard to perform basic tasks, such as choosing menu options and buttons
- Work with windows, such as selecting, minimizing, restoring, maximizing, sizing, scrolling, closing, and so forth
- Work with dialog boxes

If these tasks are unfamiliar, then refer to your Windows online documentation or complete an online Windows tour.

Manual conventions

When using this manual, note the following conventions.

Convention	Meaning/Use	Examples
bold	Used for names of options, buttons, fields, and other items that appear on the screen.	OK is a button on the screen. Click OK , or press Enter.
angle bracket (>) between items	Shows the menu option(s) that must be selected, in sequence, to get to a specific option.	From the Searches menu, select Other Local Searches > Accident .
plus sign (+) between keys	Used for keys that are pressed at the same time. Hold down the first key, and then press the other key(s). When a keystroke is available for a mouse action, both the mouse action and the keystroke are presented.	Press Ctrl+E. Click Close , or press Ctrl+F4.
comma (,) between keys	Used for keys that are pressed in sequence. Press and release each key, in the order shown.	Press Alt, F, O to open the File Options dialog box.
Courier font	Used for displayed text. Used for table names.	The software prompts: Are you sure you want to delete this record? Open the Names table (nmmain).
bold Courier font	Used for information you enter.	Enter the street address, such as 401 W Sycamore St.
<i>italics</i>	Used for emphasis. Used for variable information you supply.	Enter the date, using the <i>mm/dd/yyyy</i> format.

The following boxes indicate special information.

NOTE

Notes call attention to information that is of particular importance or that varies depending on a particular condition, such as the way your administrator (admin) has configured the software.

TIP

Tips present recommendations, optional actions, and additional ways to perform specific tasks.

CAUTION

Cautions point out actions that might endanger your data or its integrity (usefulness) or cause other problems later.

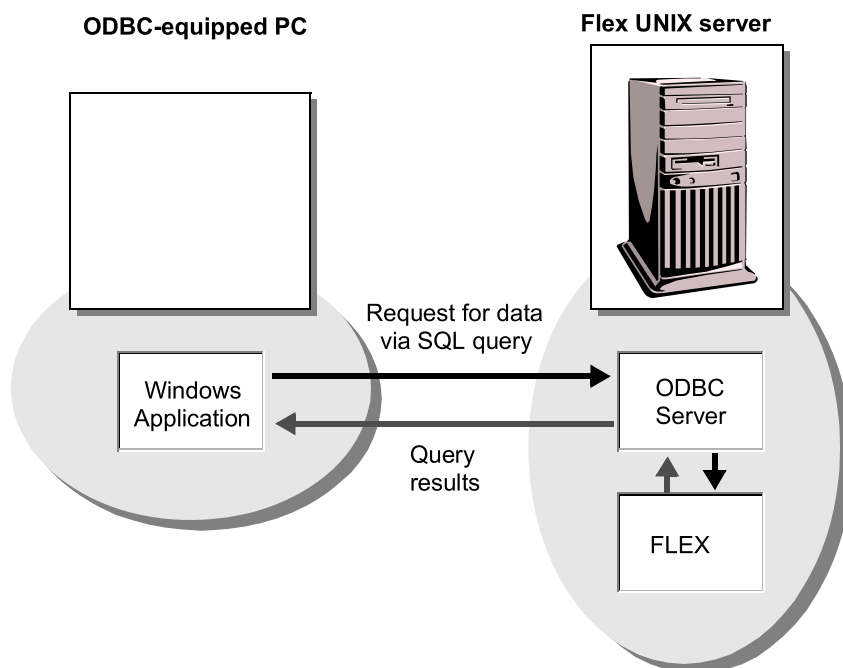
Features on your computer depend on your software version, modules, and privileges. Actual screens on your computer might vary from the example screens shown in this manual. However, any differences are minor and do not affect the tasks being described.

To find more manuals, visit the [Flex Knowledgebase](#).

Introduction

The Flex database is compliant with Open Database Connectivity (ODBC) version 3.0. This compliance allows you to access the Flex database using ODBC protocol to extract data from Flex tables and fields, via SQL queries.

The following diagram illustrates how the ODBC Interface extracts data from the Flex database.



You access the ODBC Interface from a Windows application, such as Crystal Reports® or Microsoft® Access®. This application sends an SQL query to the ODBC server located on your Flex UNIX server. The server transfers the request to Flex, which then sends the results of the query back through the ODBC server to the Windows application.

CAUTION

Although you can access the ODBC Interface using different Windows applications, Motorola Solutions provides technical support for the ODBC Interface only when it is used with Crystal Reports.

Installing the ODBC Interface

To use the ODBC Interface, you need the following hardware and software:

- PC(s) connected by a network to the Flex UNIX server
- ODBC Interface (provided by Motorola Solutions) installed on the PC(s) and the Flex UNIX server
- A Flex UNIX server that meets the operating system requirements of the third-party ODBC vendor

TIP

For exact hardware and software specifications, refer to the technical product summary (tps) for the ODBC Interface. You can obtain a copy of the tps through Technical Support.

Installation tasks

To set up the ODBC Interface, complete the following installation tasks:

- Make sure that each ODBC user has a name record in the Official Names Code table (`apnames`). (Refer to the *Flex Administrator Manual* for information on setting up `apnames` records.)
- Add a User Privileges (`sypriv`) record for each ODBC user that you want to be able to access the program `odbc`. (See [“Setting up user privileges for ODBC users” on page 9.](#))
- Add a port entry in the Services file on each PC from which you want to access the ODBC Interface. This entry must match the port number entered in the `/etc/services` file on the Flex UNIX server. (See [“Adding a port entry in the Services file” on page 10.](#))
- Add an entry for the Flex UNIX server in the Hosts file on each PC from which you want to access the ODBC Interface. For example, if your Flex UNIX server is named *motorola* and its IP address is 190.3.15.21, enter the following information in the Hosts file on your PC:

```
190.3.15.21      motorola
```

See [“Checking the Hosts file” on page 11.](#)

- Install the ODBC Interface on each PC from which you want to use the interface.

Setting up user privileges for ODBC users

In the User Privileges screen (`sypriv`), add a record for each user or group of users that you want to be able to access the program `odbc`.

To add a User Privileges record:

1. At the command line, enter **sypriv**. The User Privileges screen appears.

2. Click the **Add** button.
3. In the first field, enter **u**, **g**, or **w**, depending on whether the record applies to one user, a group of users, or all users (the world).
4. If the record is for a user or group of users, enter the user ID number or group ID number in the second field. Click the Lookup button (Ctrl+E) to select the ID number from the lookup list.

NOTE

If you are defining privileges for the world, the software skips the second field.

5. In the third field, enter **odbc**.

6. Enter **1** in the remaining fields, similar to the following sample.

Flex saves the record after you enter information in the last field.

Adding a port entry in the Services file

On each PC from which you want to access the ODBC Interface, add a port entry for the Flex UNIX server in the Services file.

To add a port entry in the Services file:

1. From the **Start** menu, click **Run**. Type one of the following commands, depending on your PC's operating system.

If you are using	Type
Windows 98	notepad c:\windows\services
Windows NT	notepad c:\winnt\system32\drivers\etc\services
Windows 2000 or later	notepad c:\windows\system32\drivers\etc\services

2. Click **OK** or press Enter. The operating system opens the Services file in Notepad.
3. Insert the cursor below the existing text.

4. Add a port entry, similar to the following sample:

```
sqlnw          1990/tcp          #Motorola ODBC
```

The port number must match the port number entered in the /etc/services file on the Flex UNIX server.

TIP

To determine the port number entered in the /etc/services file on the Flex UNIX server, enter **sh** at the command line. Then, enter the following command at the UNIX \$ prompt: **cat /etc/services | grep sqlnw**

5. Select **File > Save** to save your entry.
6. Select **File > Exit** to close Notepad.

Checking the Hosts file

Make sure that the Hosts file on each PC from which you want to access the ODBC Interface contains an entry for the Flex UNIX server.

To check the Hosts file on a PC:

1. From the **Start** menu, click **Run**. Type one of the following commands, depending on your PC's operating system.

If you are using	Type
Windows 98	notepad c:\windows\hosts
Windows NT	notepad c:\winnt\system32\drivers\etc\hosts
Windows 2000 or later	notepad c:\windows\system32\drivers\etc\hosts

2. Click **OK** or press Enter. The operating system opens the Hosts file in Notepad.
3. See whether the Hosts file contains an entry for your Flex UNIX server.

If no entry exists, insert the cursor below the other hosts listed in the file. Then, type the IP address and name of the Flex UNIX server. For example, if your Flex UNIX server is named *motorola* and its IP

address is 190.3.15.21, enter the following information in the Hosts file:

190.3.15.21 motorola

4. Select **File > Save** to save your entry.
5. Select **File > Exit** to close Notepad.

CAUTION

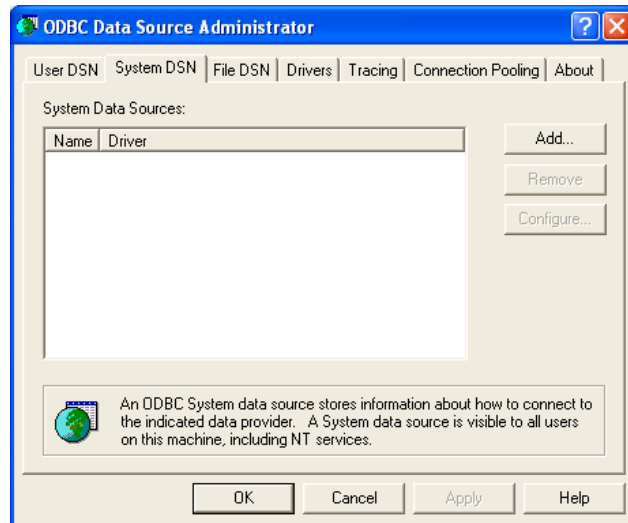
Occasionally, when you modify the Hosts or Services file, Notepad changes the file to a text document and adds the .txt extension to the file name. The .txt extension makes the file inoperable. To check the file type assigned to the Hosts or Services file, open the directory that contains the file. Right-click the file and select **Properties** from the shortcut menu. In the dialog box that appears, the **Type of File** field should display **File**. If the field displays **Text Document**, contact Technical Services for help fixing the file type.

Installing ODBC Interface on a PC

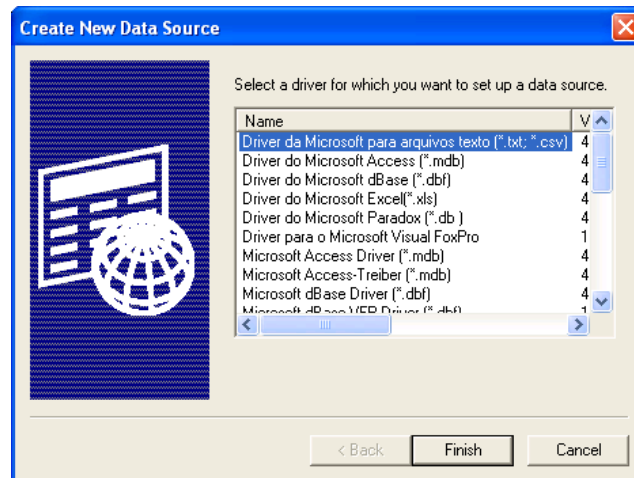
To install the ODBC Interface on a PC connected by a network to the Flex UNIX server, follow these steps:

1. Install the drivers provided by Technical Services.
2. After you install the ODBC drivers, open the Control Panel and locate the icon for ODBC setup. The icon might be named **Data Sources (ODBC)** or **ODBC Data Sources**.
3. Double-click the ODBC icon. The **ODBC Data Source Administrator** dialog box appears.

4. Click the **System DSN** tab.



5. Click the **Add** button. The **Create New Data Source** dialog box appears.



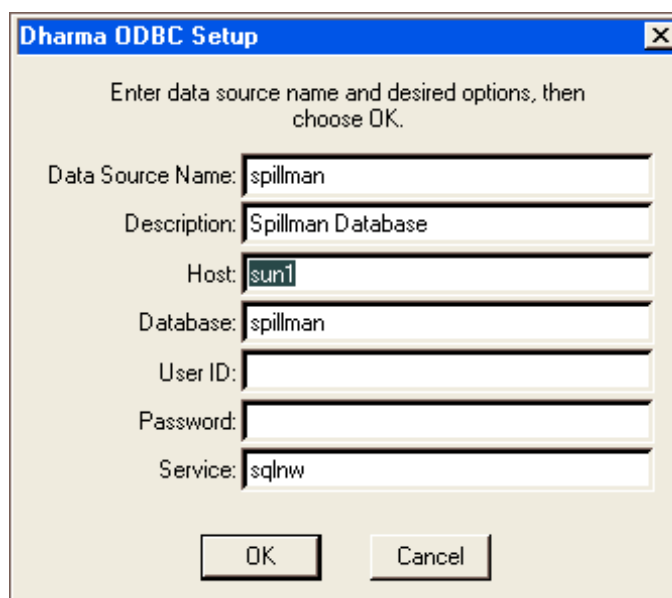
6. In the list of ODBC drivers, double-click **Dharma Datalink SDK**. (You might need to use the scroll bar to locate **Dharma Datalink SDK**.) The **Dharma ODBC Setup** dialog box appears.
7. In the **Data Source Name** field, enter **spillman** (the name of the ODBC data source—the database name created in the \$ODBCDIR directory).
8. In the **Description** field, enter a description of the ODBC data source, for example, **Motorola Database**.

9. In the **Host** field, enter the host name of the Flex UNIX server on which the ODBC daemon is running, for example, **sun1**.
10. In the **Database** field, enter **spillman** (the database name created in the \$ODBCDIR directory). The software is case sensitive, so do not capitalize **spillman**.

NOTE

You can enter your UNIX username and UNIX password in the **User ID** and **Password** fields. However, Motorola Solutions recommends that you leave these fields blank so that the software prompts you for a username and password each time you run a report. This allows more than one person to run the reports.

11. In the **Service** field, enter **sqlnw** (unless directed by Technical Services to enter a different value).



12. Click **OK**.
13. Click **OK** again to close the **ODBC Data Source Administrator** dialog box.
14. If you are using Crystal Reports, install the registry patch provided by Technical Services.

Initiating an SQL Query Through the ODBC Interface

Because of the differences in various Windows applications, this manual cannot describe exactly how to initiate an SQL query through the ODBC Interface. However, the following steps might help you understand how to initiate a query in your Windows application.

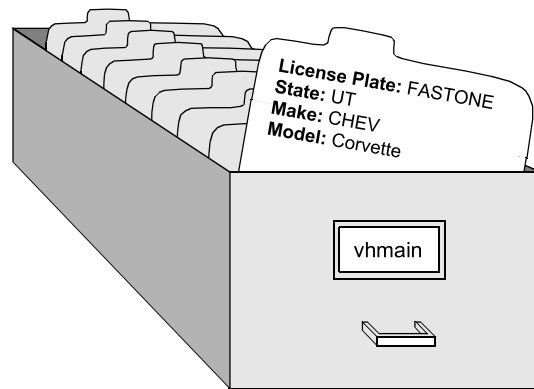
In your Windows application, you might need to:

1. Choose to create a report.
2. Indicate that the data for the report is coming from an SQL ODBC query.
3. Log on to the Flex ODBC server from your Windows application, using your UNIX username and password.
4. Set up report options, such as the date/time and narrative options. Some report writers let you choose how the data appears.
5. Create relationships between various fields and tables in Flex. Some report writers automatically do this for you; others do not. Depending on your specific application, this process might vary. To help you understand the kind of relationships you might want to identify, see [“Identifying field relationships—an example” on page 25](#).
6. Select specific Flex tables and fields from which your report displays data. Read this manual carefully to understand how to decide on these tables and fields.
7. After determining the tables and fields to include in your report, use standard SQL language structure to query the database.

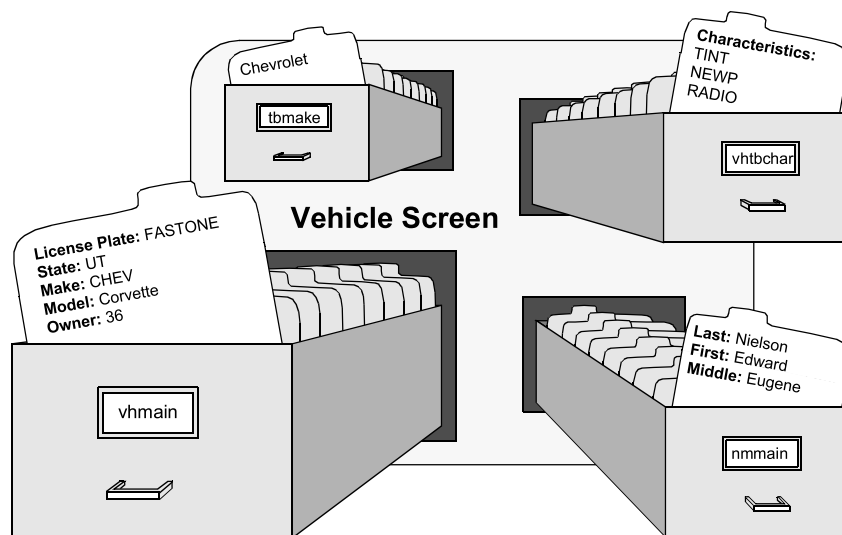
Understanding which tables and fields to include on the report

Flex screens frequently display information that is actually stored in several different tables. When you extract information from Flex through the ODBC Interface, you must retrieve the information from the table in which the information is actually stored.

Like a file drawer, each table is a collection of a particular type of records. For example, the Vehicle records, which contain most of the information you see on the Vehicle screen, are stored in the `vhmain` table. Each record contains various pieces of information, with each piece held in a field. For example, the license plate number is held in the **License Plate** field.



If you imagine each table (whose information appears on a screen) as a file drawer, the Vehicle screen might look like this:



The distinction of where the information is stored makes a significant difference in the information you extract from Flex (and the information that appears on a custom report). To have accurate reports, you must tell the Windows application where the information is stored, not where the information appears on a Flex screen. Review the following examples:

- If you want the Owner number to appear on your report, choose (in your Windows application) to extract the `ownerid` field from `vhmain`.
- If you want the first and last name of the owner to appear on your report, you must perform the following tasks in your Windows application:
 - Make sure that a relationship exists between the `ownerid` field in `vhmain` and the `number` field in `nmmain` (see [“Identifying field relationships—an example” on page 25](#) for more information).
 - Select the `first` and `last` fields from the referenced table, `nmmain`, to appear on your report.
- If you want the abbreviation of the vehicle make (for example, CHEV) to appear on the report, simply choose (in your Windows application) to extract the `make` field from `vhmain`.
- If you want the full description of the vehicle make (for example, Chevrolet) to appear on your report, you must perform the following tasks in your Windows application:
 - Make sure that a relationship exists between the `make` field in `vhmain` and the `abbr` field in the `tmake` table. (See [“Identifying field relationships—an example” on page 25](#) for more information.)
 - Choose the `desc` field from the `tmake` code table to appear on your report.

Make sure that you select tables and fields where the information is stored, not where the information is displayed.

Determining table and field names

Selecting the right field name might require some research. The field names that appear on the screen are different from the field names used in the schema, although both names refer to the same field. When you transfer information, you use the schema field names, not screen field names.

Use the following methods to find the names of the tables and fields that you want to include in your report:

- **Flex field help.** The field help feature displays the schema table and (sometimes) the field name from the screen on which the information appears. If the field help does not list the field name, you can

sometimes open the appropriate table and check the field help there for that field name. If you are unable to open a code table because it does not exist for viewing, run the `rpschema` or `rpscreen` report instead. See [“Using field help to determine table and field names” on page 18](#).

- **The `rpschema` report.** This report lists all the field names for the table you specify. You must determine which field name listed on the report corresponds with which screen field name. You can compare the report to the field help to determine a specific field name. See [“Printing the `rpschema` report to determine table and field names” on page 20](#)
- **The `rpscreen` report.** This report lists all the field names for the table you specify, as well as all the field information for tables to the specified table. As with the `rpschema` report, you must determine which field name listed on the report corresponds with which screen field name. You can compare the report to the field help to determine a specific field name. See [“Printing the `rpscreen` report to determine table and field names” on page 21](#)

Use any combination of the preceding methods to determine table and field names.

Using field help to determine table and field names

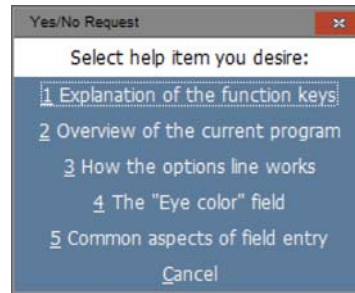
Use the Flex field help to determine table and field names. The field help lists the table and field name in the form `nmmain.eyes`, where `nmmain` is the table name and `eyes` is the field name. In most cases, this is all the information you need.

The field help might list an additional table name without a field name, for example, `Coded (from table tbcolor)`. This particular example means that additional information (which might not be visible) is pulled from the table `tbcolor`. In this case, the description of the eye color is stored in the table `tbcolor`, not in the table `nmmain`. For the eye color description to appear on the report, you must pull the information from the table `tbcolor`. To find the specific field name where that description is stored, you can run the `rpschema` report for the table `tbcolor`. (See [“Printing the `rpschema` report to determine table and field names” on page 20](#)).

To find table and field names:

1. At the command line, enter the program name for the screen that displays the information you want to include in your report. For example, to view the Names screen, enter **names**.
2. Move your cursor to the field whose table and field name you want to look up. (You can do this in Add, Modify, or Search mode.) For example, move your cursor to the **Eyes** field.

3. Press Ctrl+W to open the Help menu.



4. Select option 4 to view a description of the field. For example, select **4 The “Eye Color” field**. The **Field Description** box appears.



5. Write down the table and field names for the field.
6. Click **Ok** or press Enter to close the Help window.
7. Depending on the task you want to perform, do one of the following:
 - To get field help for another field in the same table, repeat steps 2–6 for each field needed.
 - To get field help from another table, click **Cancel** (Alt+C) and then click the **Exit** button to close the screen. Then, repeat steps 1–6.
 - If you are finished using the field help, click **Cancel** (Alt+C) and click the **Exit** button to close the screen.

Printing the rpschema report to determine table and field names

The System Schema report (`rpschema`) lists each field in the specified table and the formatting information for that field. The schema can include more fields than actually appear on a screen. Some fields are system generated. Others provide background information.

When you create a report, you can include any fields that appear on the schema (even if the information does not appear on any screen). For example, the fields `modwhen` and `modby`, which do not appear on the Names screen, indicate who last modified the information and the date the modification occurred. When using the `rpschema` report, determine which field name listed on the report corresponds with which screen field name. Use the field help to verify a field name.

To print a table schema:

1. Enter **`rpschema`** at the command line. The System Schema Reports screen appears.

The screenshot shows the 'rpschema - System Schema Report: Data Dictionary Report by Table Name' window. On the left, a list of reports is displayed, with 'rpschema - System Schema Report' selected. On the right, the 'Search Criteria' section is visible, containing a 'Table name' field with a 'Type' button, and other fields for 'Variable name', 'Description of variable', 'Data type of variable', 'Display size of field', and 'Key this references'. The 'Table name' field is highlighted. At the bottom, there are buttons for 'Close', 'Print...', and 'Recall', and a status bar showing 'OVR'.

2. In the **Table name** field, enter the name of the table from which you need information.
3. Click **Print** (Alt+P).

The **Print** dialog box appears.

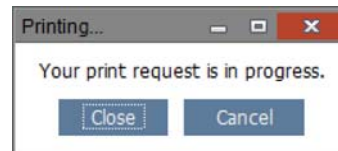
4. Set your printing options. See the *RMS User Manual* for more information.

NOTE

If you want to view the report information before printing, click the **Preview** button.

5. Click **Print** (Alt+P).

The following dialog box appears.



After processing the information, Flex removes the dialog box and sends the information to the selected destination. To close the dialog box before the information is processed, click **Close** or press Enter. To cancel the print request, click **Cancel**. (If you click **Cancel**, the software notifies you that the print job has been canceled.)

6. To print another table schema, repeat steps 2–6.
7. To close the System Schema Reports screen, click the **Exit** button.

Refer to [“Studying the rpschema and rpscreen reports” on page 23](#) for tips on understanding the information generated by the `rpschema` report.

Printing the rpscreen report to determine table and field names

The Screen Schema report (`rpscreen`) lists the formatting information for a particular screen, including the schema for each table from which a screen pulls information. For each table, the schema lists each field and the formatting information for those fields.

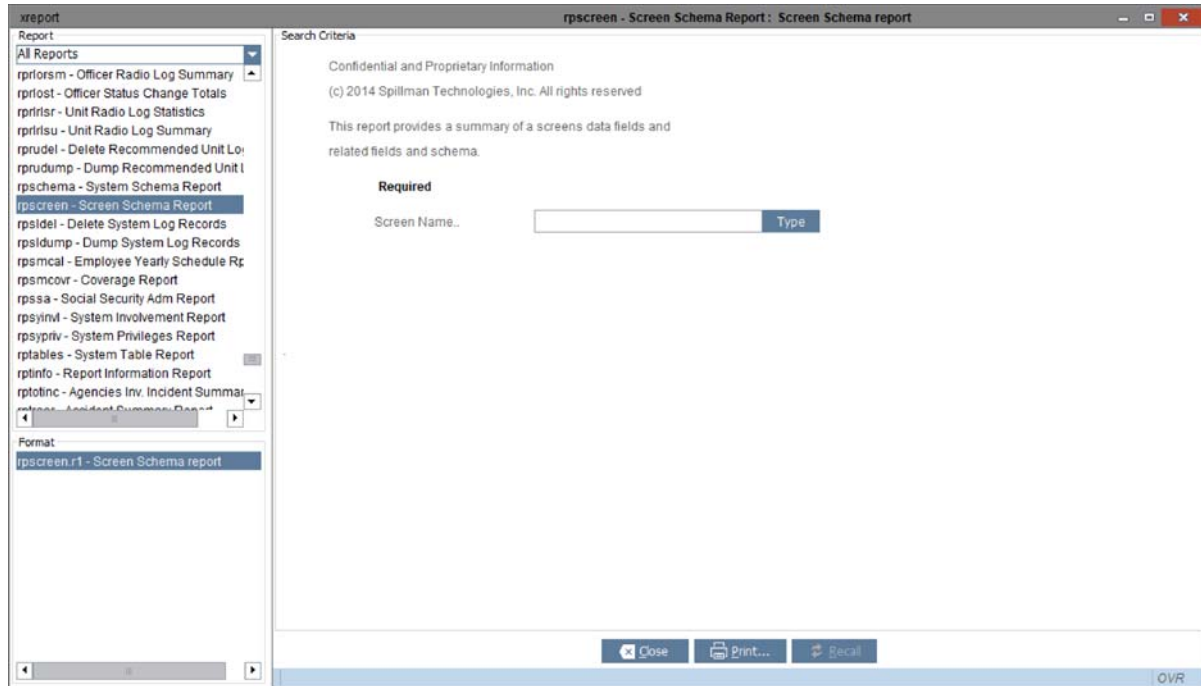
The schema can include more fields than actually appear on a screen. Some fields are system generated. Others provide background information.

A schema report can include any field that appears on the schema (even if the information does not appear on any screen). For example, the fields `modwhen` and `modby`, which do not appear on the Names screen, indicate who last modified the name information and when the modification occurred.

Determine which field name listed on the report corresponds to which screen field name. You can use the field help to verify a field name.

To print a Screen Schema report:

1. Enter **rpscreen** at the command line. The Screen Schema Report screen appears.



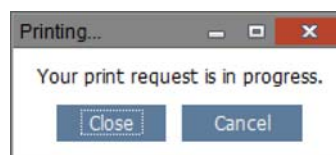
2. In the **Screen Name** field, enter the name of the primary table for a screen (not the screen name). For example, enter **nmmain** for the Names screen.
3. Click **Print** (Alt+P).
The **Print** dialog box appears.
4. Set your printing options. See the *RMS User Manual* for more information.

NOTE

If you want to view the report information before printing, click the **Preview** button.

5. Click **Print** (Alt+P).

The following dialog box appears.



After processing the information, Flex removes the dialog box and sends the information to the selected destination. To close the dialog box before the information is processed, click **Close** or press Enter. To cancel the print request, click **Cancel**. (If you click **Cancel**, the software notifies you that the print job has been canceled.)

6. To run another report, repeat steps 2–5.

7. To close the Screen Schema Report screen, click the **Exit** button.

Refer to [“Studying the rpschema and rpscreen reports” on page 23](#) for tips on understanding the `rpscreen` report.

Studying the rpschema and rpscreen reports

After you print the `rpschema` or `rpscreen` reports that you want, assess each field for the formatting information you need.

TIP

When you review the `rpscreen` report, skip to the Schema section. You do not need the first few pages of the report.

The following report shows the `rpscreen` report for `nmmain`. The `rpscreen` report includes a description of all the fields in the `nmmain` table as well as a description of any other tables linked to `nmmain`, such as

nmalert. (The rpschema report for nmmain is similar to the rpscreen report, except the rpschema report does not include the schema information for nmalert or other tables linked to nmmain.)

```
09/18/17                                COLLEGE STATION POLICE                                500
16:51                                Screen/Schema Report                                Page: 1
Confidential/Proprietary (c)2017 Spillman Technologies, Inc. All rights reserved

Var. Name      Table      Ref. Key      Type      Len.
-----
nmalert.code   nmalert  nmtbalrt.abbr String     4
nmalert.number nmalert  nmmain.number String     9
nmalert.seq    nmalert  nmmain.number Short      3

Schema Section:
nmalert
-----
Seq Field      Ref. Key      Type      Size Desc
-----
1  number      nmmain.number String     9  Name number
2  seq          nmmain.number Short      3  Sequence number
3  modwhen      nmmain.number xtime     19  When Modified
4  modby        nmmain.number Modified By 5
5  code         nmtbalrt.abbr String     4  Alert code for above name

Report Includes:
Table Equal to Table nmalert
```

CAUTION

The preceding sample report might not contain current information. The report is only an example.

Use the following table to help you understand the sample report.

Area of the report	Definition
Table name	The name of the table whose schema follows
Field name	The name of a field in the table
Reference key	The table and field name where the list of codes resides for coded or referenced fields
Description	A description of the field

The `rpscreen` report helps you gather the following information about the fields on the report:

- The type of information that you find in each field (for example, a first name in the **First** field). Knowing this can help you decide what you want to appear on your report.
- The table and field names for the fields that you want to appear on your report.
- The table and field names from which coded fields pull information. Knowing this helps you identify the relationships between fields in your Windows application.

Identifying field relationships—an example

The following example shows how you might determine which table and field relationships to identify in your Windows application and which fields to include in the report. Suppose that you want to create a report from the Names table, and you want the report to include a description of the hair color, not the hair code.

To create the report:

1. Identify (in your Windows application) that a relationship exists between the `hair` field in `nmmain` and the `abbr` field in `tbcolor`. You know that this is the correct relationship because the Reference Key (found in the `rpscreen` or `rpschema` report) for the `hair` field in `nmmain` is `tbcolor.abbr`.
2. Print the `rpschema` for the table `tbcolor`. (See [“Printing the rpschema report to determine table and field names”](#) on page 20.)
3. Determine from the `rpschema` that the description of the hair color is held in the field `desc`.
4. Select (in your Windows application) the `desc` field from `tbcolor` to appear on your report.

Using the ODBC Log

The ODBC log tracks which users have accessed Flex through the ODBC Interface and whether any “fatal” errors have occurred. This log is found in the root directory of the release, for example, `$ODBCDIR/sql_server.log`. You can turn the logging process on or off by setting the `odbclog` application parameter.

To set the `odbclog` application parameter:

1. Enter **apparam** at the command line to open the Application Parameters screen.

2. click the **Srch** button.
3. In the **Application Parameter Name** field, type `odbclog`.
4. Click **Accept** (Alt+A).
 - If the `odbclog` application parameter already exists, display the record and click the **Mod** button. Then, go to the next step.
 - If the application parameter does not exist, follow these steps to add the parameter.

To add the `odbclog` application parameter:

1. Click the **Add** button.
2. In the **Application Parameter Name** field, enter `odbclog`.
3. In the **Parameter Description** field, type a description of the application parameter. Then, go to step 5.
5. In the **Application Parameter Value** field, type **YES** to turn on logging or type **NO** to turn off logging.

6. Click **Accept** (Alt+A).

7. Click the **Exit** button.

CAUTION

If the ODBC file becomes too large, delete the file. ODBC generates a fresh ODBC log.

Controlling the ODBC daemon

The ODBC daemon, located on the Flex UNIX server, controls data passing through the ODBC Interface. For requests to be received and results to be sent, the ODBC daemon must be running.

You can use the following commands to start, stop, and check the status of the ODBC daemon.

To	Enter this command at the UNIX prompt
Start the ODBC daemon	Dstart -Co
Stop the ODBC daemon	Dstop -Co
Check the status of the ODBC daemon	Dstat -Co

NOTE

You can start or stop the ODBC daemon only if you are logged on to Flex as root, sds, or odbc. All Flex users can use the `Dstat -Co` command to check the status of ODBC daemon.

For example, you might use the `Dstart -Co` command if you have stopped all the Flex daemons and want to manually restart the ODBC daemon.

To perform an ODBC command, enter **sh** at the command line. At the UNIX \$ prompt, enter the ODBC command.

Understanding Error Messages

Refer to this section for information about Flex database errors. Error messages that are not listed here are ODBC server errors or Windows application errors. If you are not sure how to resolve the situation, contact Technical Services.

- Error 1001** Not yet implemented error.
- This error appears if you attempt an action that has not been implemented in the Flex code. For example, the Flex code does not allow for an update of the Flex database through ODBC. Only read-only access is allowed.
- Error 1002** No Such Table.
- This error appears if you try to access a table that is not found in the daemon server's data dictionary. Generally, this error occurs because the table name was missing the "sds" prefix. All table names use the following format: `sds.table_name`. Crystal Reports automatically adds the "sds" prefix to all table names. Unless the user modifies the query manually, this error message should not occur in Crystal Reports.
- Error 1003** Invalid offset error.
- This error appears if an error occurred during the calculation of a variable length record. If this error occurs, the code might have a bug or the database might be corrupted.
- Error 1004** An invalid operator was sent to Flex for an index fetch.
- This error might appear if the Dharma or Flex code has a bug.
- Error 1005** Specified index is not found.
- Flex cannot find the index that the Dharma server picked. The Dharma data dictionary is out of sync with Flex.
- Error 1008** Invalid login or password.
- Permission denied to Flex. This error might appear for any of the following reasons:
- You enter a username and password that the `/etc/passwd` file cannot verify.
 - You do not have an entry in `apnames`.

- You do not have a `sypriv` record that gives you access to the program `odbc`.

